MINI PROJECT REPORT

AI-POWERED CYBERSECURITY

Account Security, Phishing Detection, Sign-in Alert Analysis and Evaluation of AI-Generated Security Advice

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ABSTRACT

Cybersecurity plays an important role in protecting digital accounts, personal information, devices, and online services from unauthorised access and cyberattacks. This mini project focuses on four important areas of cybersecurity: student account security, phishing email detection, sign-in alert investigation, and evaluation of AI-generated security advice.

The project uses fictional scenarios and synthetic login records to understand common security risks and identify suitable preventive measures. It explains the importance of strong passwords, multi-factor authentication, software updates, data backups, safe email verification, and proper incident response.

The project also demonstrates how AI-generated security advice can be evaluated critically to identify incorrect assumptions and unsafe recommendations. The main

objective is to develop cybersecurity awareness and promote safe digital practices among students.

1. INTRODUCTION

Cybersecurity refers to the methods and technologies used to protect computers, networks, digital accounts, and information from unauthorised access, damage, theft, and misuse.

Nowadays, students use online learning portals, email accounts, cloud storage, and digital platforms for educational activities. These platforms may be exposed to threats such as phishing, password theft, malicious software, and unauthorised login attempts.

Cybersecurity helps protect the confidentiality, integrity, and availability of information.

The three main principles of cybersecurity are:

1. Confidentiality: Ensures that information is accessible only to authorised users.
2. Integrity: Ensures that information remains accurate and is not changed without permission.
3. Availability: Ensures that information and services remain accessible when required.

This mini project studies common cybersecurity problems through fictional examples and recommends practical methods to prevent and respond to security incidents.

2. PROBLEM STATEMENT

Students frequently use digital platforms for communication, learning, and storing academic information. However, unsafe practices such as reusing passwords, approving unexpected login requests, delaying software updates, and clicking suspicious links can increase the risk of cyberattacks.

Students may also find it difficult to identify phishing emails, understand unusual login activities, and evaluate security advice generated by artificial intelligence.

Therefore, there is a need to study these security risks and develop practical recommendations to improve account protection, phishing awareness, and incident response.

3. AIM AND OBJECTIVES

3.1 Aim

To study fundamental cybersecurity risks and develop practical solutions for student account protection, phishing detection, sign-in alert analysis, and safe use of AI-generated cybersecurity advice.

3.2 Objectives

1. To identify common risks associated with student accounts.
2. To understand the importance of strong and unique passwords.
3. To identify warning signs in suspicious emails.
4. To study login records and recognise potentially unauthorised access.
5. To understand the importance of multi-factor authentication.
6. To evaluate incorrect or unsafe AI-generated security recommendations.

7. To suggest appropriate cybersecurity controls and incident-response procedures.
8. To improve awareness of safe online practices.

4. SCOPE OF THE PROJECT

The scope of this project includes:

1. Student account security and password management.
2. Identification of phishing indicators in suspicious messages.
3. Analysis of fictional sign-in logs.
4. Understanding authentication and authorisation.
5. Evaluation of AI-generated cybersecurity advice.
6. Recommendations for protecting accounts and digital information.
7. Basic cybersecurity awareness and incident response.

The project is based on fictional scenarios and synthetic evidence provided in the EWB Assessment 01 assignment. It does not involve scanning public systems, testing live websites, or accessing real user accounts.

5. METHODOLOGY

The project follows a scenario-based study methodology.

Step 1: Study of cybersecurity risks

The supplied student-account scenario is examined to identify security weaknesses and possible consequences.

Step 2: Risk assessment

Each identified risk is analysed according to the confidentiality, integrity, and availability principles.

Step 3: Phishing analysis

A fictional scholarship email is examined to identify suspicious language, requests, and sender details.

Step 4: Sign-in alert investigation

Synthetic login records are arranged chronologically to identify unusual authentication activity and possible account compromise.

Step 5: Evaluation of AI advice

The supplied AI-generated security statements are reviewed to identify unsupported claims and unsafe actions.

Step 6: Recommendations

Practical security controls and incident-response measures are suggested based on the available evidence.

6. TASK A — STUDENT ACCOUNT SECURITY

6.1 Case Study

In the given scenario, a student named Mira uses the same short password for her email account and learning portal. She approves unexpected sign-in prompts, delays device updates, stores her only project copy on one laptop, and allows a friend to use the laptop while her account remains signed in.

These practices can expose her accounts and academic information to several security risks.

6.2 Identification of Security Risks

Risk	Asset at Risk	CIA Impact	Security Control	Limitation
Weak and reused password	Email and learning portal	Confidentiality and integrity	Use a long, unique password for each account.	A strong password cannot prevent every phishing attack.
Approving unexpected MFA prompts	Online accounts	Confidentiality and integrity	Deny unexpected prompts and report repeated requests.	Users may still be tricked into approving requests.
Delaying software updates	Laptop and stored files	Confidentiality, integrity and availability	Install security updates promptly.	Updates cannot prevent every cyberattack.
Keeping only one project copy	Academic files	Availability	Maintain a separate backup.	Backups may be outdated or inaccessible.
Sharing a signed-in laptop	Personal information and account sessions	Confidentiality and integrity	Lock the device and use separate user accounts.	Protection depends on consistent use.

6.3 Prioritised Security Actions

1. Secure account credentials

Change weak or reused passwords and create unique passwords for each account. Deny unexpected MFA prompts. These actions reduce the risk of unauthorised account access.

2. Protect the device

Install pending security updates, enable a screen lock, sign out after use, and avoid allowing others to access a signed-in account.

3. Create data backups

Keep an additional copy of academic files in an approved cloud service or on an external storage device. Check that the backup can be restored when needed.

6.4 Authentication and Authorisation

Authentication: Authentication verifies the identity of a user. For example, Mira enters her learning portal credentials and completes an approved second authentication step.

Authorisation: Authorisation determines what an authenticated user is allowed to access or modify. For example, Mira may view course materials and submit assignments, but she may not have permission to change administrator settings.

7. TASK B — PHISHING EMAIL DETECTION

7.1 Introduction

Phishing is a cyberattack in which an attacker attempts to trick users into revealing confidential information, clicking malicious links, or performing unsafe actions.

In the given scenario, a fictional scholarship email claims that a scholarship will expire within 20 minutes. It asks the student to enter a college password and OTP and send a photograph of an identity card.

The message contains several warning signs that require careful investigation.

7.2 Identification of Warning Signs

Warning Sign	Explanation
“ FINAL WARNING”	Urgent language may pressure the recipient into making a quick decision.
“ expires in 20 minutes”	A short deadline can discourage careful verification.
Different sender and Reply-To addresses	The difference between the displayed sender domain and reply-to domain should be investigated.
“ enter your college password and OTP”	Requesting passwords and OTPs is a serious security warning sign.
“ Do not contact the college office”	Discouraging independent verification may be an attempt to prevent the recipient from discovering fraud.
“ Reply with a photo of your identity card”	Sending identity documents through an unverified channel may expose sensitive personal information.

7.3 Safe Verification Process

1. Do not click the suspicious link or reply to the message.
2. Never enter a password or OTP through an unverified link.
3. Open the college portal using its known official address or a saved bookmark.
4. Contact the scholarship office using independently obtained official contact details.

5. Report the suspicious message through the institution's approved reporting process.
6. Preserve the email and relevant information if an investigation is required.

HTTPS alone does not prove that a website is genuine. It encrypts the connection between the browser and the website, but a fraudulent website can also use HTTPS. Similarly, a familiar display name does not guarantee that a message comes from the claimed organisation.

7.4 Response Plan if Credentials Are Compromised

If a student has already entered a password and OTP:

1. Change the affected password immediately using the official service and a trusted device.
2. Change any other account passwords that were reused.
3. Revoke active sessions where the service allows it.
4. Contact the official IT support team.
5. Review recent sign-ins, account recovery details, and suspicious account changes.
6. Preserve the suspicious message and relevant timestamps.
7. Follow the institution's incident-response instructions.

Limitation: The message alone cannot establish the identity of its sender or prove that the linked website is malicious. Additional evidence, such as full email headers and independent confirmation from the college, would be needed.

8. TASK C — SIGN-IN ALERT ANALYSIS

8.1 Introduction

Sign-in alert analysis involves reviewing authentication records to identify unusual or potentially unauthorised access.

The supplied fictional log contains password attempts, multi-factor authentication events, and session creation. The events occur on 18 October 2026.

8.2 Sign-in Timeline

Time	Observed Event	Interpretation
09:00	Password success from 192.0.2.10	Consistent with normal access according to the supplied lab note.
09:01	MFA approved	The authentication process was completed.
10:14	Password failed from 198.51.100.23	An unsuccessful login attempt occurred.
10:15	Password failed again	A second unsuccessful attempt occurred.
10:16	Password success	The password was accepted, but the identity of the person using it is not established.
10:17	MFA denied	An authentication prompt was denied.
10:18	MFA approved	A prompt was approved. The student states that they approved it because they thought it would stop repeated requests.
10:20	Session created	A session was created after the authentication events.

8.3 Likely Explanation

One possible explanation is that an unauthorised person obtained or guessed the student's password and gained access after the MFA prompt was approved.

The repeated failed attempts followed by a successful password event, an MFA denial, an MFA approval, and session creation create concern about possible account compromise.

However, the available evidence does not establish the identity of the person using the source IP address.

8.4 Alternative Explanation

An alternative possibility is that the activity resulted from a legitimate service, network route, VPN, or another account-use circumstance that is not represented in the limited log.

Further investigation is required before reaching a definite conclusion.

8.5 Additional Checks

1. Review the complete authentication and session logs, including available device information, MFA method, and subsequent account activity.
2. Confirm the event timeline with the student and investigate whether a known device, VPN, or institutional service could explain the activity.

8.6 Recommended IT Response

Priority 1: Protect the account

Revoke the suspicious session, reset credentials through the official process, review recovery settings and MFA registrations, and require fresh authentication.

Priority 2: Preserve evidence

Retain the relevant logs, timestamps, original alert, and user statement. Document the actions taken during the investigation.

Priority 3: Investigate and monitor

Review recent account changes and activity, check for similar events involving other accounts, and follow the institution's approved incident-response procedure.

8.7 Incident Summary

The synthetic sign-in log shows two failed password attempts followed by a successful password event, an MFA denial, an MFA approval, and session creation. The student reports not initiating a sign-in after 09:05 and states that an MFA prompt was approved at approximately 10:18. This pattern raises concern about possible account compromise, but the available evidence is insufficient to identify a person or establish how access occurred. The authorised IT team should secure the account, revoke the suspicious session, preserve the logs, and investigate the authentication activity.

9. TASK D — EVALUATION OF AI-GENERATED SECURITY ADVICE

9.1 Introduction

Artificial intelligence can help analyse cybersecurity scenarios, summarise evidence, and suggest preventive actions. However, AI-generated information can contain inaccurate statements, unsupported conclusions, or unsafe recommendations.

Therefore, AI output should be checked against the available evidence and trustworthy cybersecurity guidance.

9.2 Identification of Unsafe Statements

AI-Generated Statement	Correction
“ The email is safe because it uses HTTPS.”	HTTPS does not establish that the website or message is genuine. The recipient should verify the message independently.
“ The logs prove that the person using 198.51.100.23 stole the password.”	The logs record events associated with an example IP address. They do not establish the identity of a person or prove how the password was obtained.
“ MFA prevents every account compromise.”	MFA reduces risk but does not eliminate it. Users can still be tricked into approving unexpected prompts.
“ Delete the suspicious logs. ”	Relevant logs should be preserved as evidence and handled through the authorised incident-response process.
“ Send the student's password to support so they can test it.”	Passwords and OTPs must not be shared. IT teams should use approved recovery and investigation procedures.

9.3 Improved Prompt for AI Analysis

The following prompt can be used to request a safer analysis:

“ Analyse only the fictional cybersecurity evidence provided. Separate directly observed facts from interpretations and assumptions. Identify uncertainty and alternative explanations. Do not identify a person from an IP address or make claims beyond the evidence. Do not request passwords, OTPs, identity documents, or other

confidential information. Recommend safe, prioritised actions for the account holder and authorised IT team. Explain the limitations of the evidence and refer to trustworthy cybersecurity guidance where relevant.”

9.4 Corrected Cybersecurity Advice

The scholarship email should be treated as suspicious because it creates urgency, requests a password and OTP, discourages contact with the college, and asks for an identity-card photograph. The recipient should not click the link or provide the requested information. The scholarship should be verified using the official college portal or independently obtained contact details.

If credentials have already been entered, the affected password should be changed promptly, active sessions should be revoked where possible, and the incident should be reported to official IT support.

The sign-in log shows a concerning sequence of authentication events, but it does not identify an individual or prove how access was obtained. The authorised IT team should preserve the evidence, protect the account, and investigate further. HTTPS and MFA are useful security protections, but neither guarantees complete safety.

10. RESULTS AND DISCUSSION

The project identifies several important cybersecurity risks and appropriate methods for reducing them.

The student-account analysis demonstrates the importance of using unique passwords, handling MFA prompts carefully, installing updates, and maintaining backups.

The phishing analysis shows that urgency, requests for credentials, mismatched sender details, discouragement of independent verification, and requests for identity documents are warning signs that require further scrutiny.

The sign-in alert analysis demonstrates that authentication logs can help reconstruct a sequence of events. However, a source IP address alone cannot prove who performed an action. Investigations should consider alternative explanations and preserve relevant evidence.

The evaluation of AI-generated security advice demonstrates that AI tools can produce incorrect or unsafe recommendations. Their output must be critically reviewed before being used to make security decisions.

Overall, the project highlights the importance of prevention, user awareness, independent verification, and evidence-based incident response.

11. CONCLUSION

This mini project studied foundational cybersecurity concepts through four areas: student account security, phishing email detection, sign-in alert analysis, and evaluation of AI-generated security advice.

The project identified common account-security weaknesses and proposed controls such as unique passwords, careful MFA approval, timely software updates, and separate backups. It also explained how suspicious emails can be verified safely and how authentication logs can be used to investigate unusual activity.

The analysis showed that security decisions should be based on available evidence rather than assumptions. AI-generated advice should also be reviewed to identify incorrect claims and unsafe actions.

In conclusion, effective cybersecurity requires a combination of technical controls, user awareness, careful evidence handling, and appropriate incident-response procedures.

12. FUTURE SCOPE

The project can be extended in the following ways:

1. Develop a cybersecurity awareness programme for students.
2. Create a checklist for identifying suspicious emails.
3. Develop an offline demonstration tool for visualising synthetic sign-in logs.
4. Study password managers and phishing-resistant authentication methods using published guidance.
5. Prepare a standard incident-reporting template for educational institutions.
6. Explore how AI can support cybersecurity analysis while maintaining human verification and privacy.
7. Conduct authorised cybersecurity awareness surveys to evaluate students' understanding of common threats.

13. REFERENCES

1. EWB Courses, Assessment 01: Security Foundations, AI-Powered Cybersecurity, issued 18 October 2026. Supplied assignment document.

2. Cybersecurity and Infrastructure Security Agency (CISA), Secure Our World. Available at: https://www.cisa.gov/secure-our-world

3. National Institute of Standards and Technology (NIST), Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile, NIST AI 600-1. Available at: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

14. AI-USE DECLARATION

AI tool used: ChatGPT.

Purpose: To assist in organising the supplied EWB cybersecurity assessment into mini-project format and drafting explanatory sections.

The final report should be reviewed, understood, and edited by the student to ensure that it reflects their own understanding and complies with the course's submission requirements.

Student Name: _____

Batch / Group: _____

Date: _____